

The empirical recurrence for OEIS A235732: recovering a conjecture that is not written down

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Abstract

OEIS A235732 records “Empirical recurrence of order 86”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 420$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 86, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 86 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235732 is “Number of $(n+1) \times (4+1)$ 0..1 arrays with the sum of each 2×2 subblock maximum and minimum lexicographically nondecreasing columnwise and nonincreasing rowwise.”. It has offset 1 and begins

746, 15963, 337412, 7041734, 146785037, 3054397001, 63554803821, 1322113787590, ...

The entry states:

Empirical recurrence of order 86 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:44:45, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 508$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 420$ of the 508, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 420$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1016$ exact terms of the model returns order 86 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{86} c_i a(n-i),$$

c_1	21	c_{23}	−64845994930048720	c_{45}	1911965698604773417799	c_{67}	−47124166300
c_2	218	c_{24}	65798010368174800	c_{46}	−1434424111793364587110	c_{68}	125089214402
c_3	−4812	c_{25}	372087737153374130	c_{47}	−1937959783478367242584	c_{69}	80684600132
c_4	−17546	c_{26}	−370951937240648293	c_{48}	1382994985871938725421	c_{70}	−1885797685
c_5	475588	c_{27}	−1794876903776626787	c_{49}	1692872534115705955917	c_{71}	−1129868995
c_6	645009	c_{28}	1752182988153168772	c_{50}	−1139814371262075560889	c_{72}	23402842263
c_7	−27266389	c_{29}	7317474946131156650	c_{51}	−1274168364689299361205	c_{73}	1275272343
c_8	−7290424	c_{30}	−6983269567529315920	c_{52}	802086927084217941974	c_{74}	−236785069
c_9	1034663402	c_{31}	−25321648925643227600	c_{53}	825662747730432488426	c_{75}	−113796640
c_{10}	−332123316	c_{32}	23604086673170159017	c_{54}	−481209764923057274025	c_{76}	192551995
c_{11}	−28038261364	c_{33}	74634561774343441317	c_{55}	−459933479776334140163	c_{77}	781624819
c_{12}	18389468290	c_{34}	−67914510833180179590	c_{56}	245672667508370539235	c_{78}	−12306119
c_{13}	569788647300	c_{35}	−187903532775412377772	c_{57}	219733271070539541853	c_{79}	−39730119
c_{14}	−482934056016	c_{36}	166760524055142648866	c_{58}	−106485542710361403286	c_{80}	59592911
c_{15}	−8979344185752	c_{37}	405023219985015890704	c_{59}	−89744821030830410792	c_{81}	14030137
c_{16}	8560350336900	c_{38}	−350032835678514610049	c_{60}	39079440278691173052	c_{82}	−2053912
c_{17}	112401933264324	c_{39}	−748853141920485629095	c_{61}	31204809813372693404	c_{83}	−306306
c_{18}	−113087178816384	c_{40}	628704870670322286716	c_{62}	−12104606724441224627	c_{84}	447513
c_{19}	−1137696442072132	c_{41}	1189485207405032974982	c_{63}	−9188985241666079333	c_{85}	310272
c_{20}	1167355564313859	c_{42}	−966763334896353164762	c_{64}	3153008827223332472	c_{86}	−46080
c_{21}	9437916632925649	c_{43}	−1625222973644767573546	c_{65}	2277123857980901446		
c_{22}	−9684784939045898	c_{44}	1272816926496162386527	c_{66}	−687848535435851014		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 86, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $86 + 4$ terms removed returns order 86 as well, so $\deg q_{\min} = 86 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $86 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 86 that this sequence satisfies — and that family turns out to have exactly one member.

References

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